Lab 8 Handling Text Files part 1

Question 1

Practice 1

```
pccc@cis106:~$ tac /etc/passwd
vboxadd:x:997:1::/var/run/vboxadd:/bin/false
_flatpak:x:111:115:Flatpak system-wide installation helper:/nonexistent:/usr/sbin/nologin
pccc:x:1000:1000:Mo,,,:/home/pccc:/bin/bash
Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false
colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
saned:x:107:111::/var/lib/saned:/usr/sbin/nologin
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
geoclue:x:106:110::/var/lib/geoclue:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
cups-pk-helper:x:105:108:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
speech-dispatcher:x:103:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false
avahi:x:102:107:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin
tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
games:x:5:60:games:/usr/games:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
sys:x:3:3:sys:/dev:/usr/sbin/nologin
```


```
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vboxadd:x:997:1::/var/run/vboxadd:/bin/false
_flatpak:x:111:115:Flatpak system-wide installation helper:/nonexistent:/usr/sbin/nologin
pccc:x:1000:1000:Mo,,,:/home/pccc:/bin/bash
Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false
colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
saned:x:107:111::/var/lib/saned:/usr/sbin/nologin
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
geoclue:x:106:110::/var/lib/geoclue:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
cups-pk-helper:x:105:108:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin
usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin
speech-dispatcher:x:103:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false
avahi:x:102:107:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nologin
systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin
tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false
dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
games:x:5:60:games:/usr/games:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
sys:x:3:3:sys:/dev:/usr/sbin/nologin
```

```
pccc@cis106:~$ cat -nE /etc/passwd
     1  root:x:0:0:root:/root:/bin/bash$
     2  daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin$
     3  bin:x:2:2:bin:/bin:/usr/sbin/nologin$
     4  sys:x:3:3:sys:/dev:/usr/sbin/nologin$
     5  sync:x:4:65534:sync:/bin:/bin/sync$
     6  games:x:5:60:games:/usr/games:/usr/sbin/nologin$
     7  man:x:6:12:man:/var/cache/man:/usr/sbin/nologin$
     8  lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin$
     9  mail:x:8:8:mail:/var/mail:/usr/sbin/nologin$
    10  news:x:9:9:news:/var/spool/news:/usr/sbin/nologin$
    11  uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin$
    12  proxy:x:13:13:proxy:/bin:/usr/sbin/nologin$
    13  www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin$
    14  backup:x:34:34:backup:/var/backups:/usr/sbin/nologin$
    15  list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin$
    16  irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin$
    17  _apt:x:42:65534::/nonexistent:/usr/sbin/nologin$
    18  nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin$
    19  systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin$
    20  dhcpcd:x:100:65534:DHCP Client Daemon:/usr/lib/dhcpcd:/bin/false$
    21  tss:x:101:103:TPM software stack:/var/lib/tpm:/bin/false$
    22  systemd-timesync:x:991:991:systemd Time Synchronization:/:/usr/sbin/nologin$
    23  messagebus:x:990:990:System Message Bus:/nonexistent:/usr/sbin/nologin$
    24  dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin$
    25  avahi:x:102:107:Avahi mDNS daemon:/run/avahi-daemon:/usr/sbin/nologin$
    26  speech-dispatcher:x:103:29:Speech Dispatcher:/run/speech-dispatcher:/bin/false$
    27  usbmux:x:104:46:usbmux daemon:/var/lib/usbmux:/usr/sbin/nologin$
    28  cups-pk-helper:x:105:108:user for cups-pk-helper service:/nonexistent:/usr/sbin/nologin$
    29  fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin$
    30  geoclue:x:106:110::/var/lib/geoclue:/usr/sbin/nologin$
    31  gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin$
    32  saned:x:107:111::/var/lib/saned:/usr/sbin/nologin$
    33  polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin$
    34  rtkit:x:108:112:RealtimeKit:/proc:/usr/sbin/nologin$
    35  colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin$
    36  Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false$
```

```
pccc@cis106:~$ head -n 5 /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
pccc@cis106:~$ tail -n 5 /etc/passwd
colord:x:109:113:colord colour management daemon:/var/lib/colord:/usr/sbin/nologin
Debian-gdm:x:110:114:Gnome Display Manager:/var/lib/gdm3:/bin/false
pccc:x:1000:1000:Mo,,,:/home/pccc:/bin/bash
_flatpak:x:111:115:Flatpak system-wide installation helper:/nonexistent:/usr/sbin/nologin
vboxadd:x:997:1::/var/run/vboxadd:/bin/false
pccc@cis106:~$ tail -n 1 /etc/passwd
vboxadd:x:997:1::/var/run/vboxadd:/bin/false
pccc@cis106:~$ head -n 1 /etc/passwd
root:x:0:0:root:/root:/bin/bash
pccc@cis106:~$
```

```
Tilix: pccc@cis106: ~
1: pccc@cis106: ~
15811  against accepting unsolicited donations from donors in such states who
15812  approach us with offers to donate.
15813
15814  International donations are gratefully accepted, but we cannot make
15815  any statements concerning tax treatment of donations received from
15816  outside the United States. U.S. laws alone swamp our small staff.
15817
15818  Please check the Project Gutenberg web pages for current donation
15819  methods and addresses. Donations are accepted in a number of other
15820  ways including checks, online payments and credit card donations. To
15821  donate, please visit: www.gutenberg.org/donate.
15822
15823  Section 5. General Information About Project Gutenberg™ electronic works
15824
15825  Professor Michael S. Hart was the originator of the Project
15826  Gutenberg™ concept of a library of electronic works that could be
15827  freely shared with anyone. For forty years, he produced and
15828  distributed Project Gutenberg™ eBooks with only a loose network of
15829  volunteer support.
15830
15831  Project Gutenberg™ eBooks are often created from several printed
15832  editions, all of which are confirmed as not protected by copyright in
15833  the U.S. unless a copyright notice is included. Thus, we do not
15834  necessarily keep eBooks in compliance with any particular paper
15835  edition.
15836
15837  Most people start at our website which has the main PG search
15838  facility: www.gutenberg.org.
15839
15840  This website includes information about Project Gutenberg™,
15841  including how to make donations to the Project Gutenberg Literary
15842  Archive Foundation, how to help produce our new eBooks, and how to
15843  subscribe to our email newsletter to hear about new eBooks.
15844
15845
pccc@cis106:~$
```

Practice 2

```
pccc@cis106:~$ cut -d ":" -f1 /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
dhcpcd
tss
systemd-timesync
messagebus
dnsmasq
avahi
speech-dispatcher
usbmux
cups-pk-helper
fwupd-refresh
geoclue
gnome-remote-desktop
saned
polkitd
rtkit
colord
Debian-gdm
```

```
pccc@cis106:~$ cut -d ":" -f1,6 /etc/passwd
root:/root
daemon:/usr/sbin
bin:/bin
sys:/dev
sync:/bin
games:/usr/games
man:/var/cache/man
lp:/var/spool/lpd
mail:/var/mail
news:/var/spool/news
uucp:/var/spool/uucp
proxy:/bin
www-data:/var/www
backup:/var/backups
list:/var/list
irc:/run/ircd
_apt:/nonexistent
nobody:/nonexistent
systemd-network:/
dhcpcd:/usr/lib/dhcpcd
tss:/var/lib/tpm
systemd-timesync:/
messagebus:/nonexistent
dnsmasq:/var/lib/misc
avahi:/run/avahi-daemon
speech-dispatcher:/run/speech-dispatcher
usbmux:/var/lib/usbmux
cups-pk-helper:/nonexistent
fwupd-refresh:/var/lib/fwupd
geoclue:/var/lib/geoclue
gnome-remote-desktop:/var/lib/gnome-remote-desktop
saned:/var/lib/saned
polkitd:/
rtkit:/proc
colord:/var/lib/colord
Debian-gdm:/var/lib/gdm3
```

```
PASSWD(5)                File Formats and Configuration                PASSWD(5)

NAME
       passwd - the password file

DESCRIPTION
       /etc/passwd contains one line for each user account, with seven fields
       delimited by colons (":"). These fields are:

       •   login name

       •   optional encrypted password

       •   numerical user ID

       •   numerical group ID

       •   user name or comment field

       •   user home directory

       •   optional user command interpreter

       If the password field is a lower-case "x", then the encrypted password is
       actually stored in the shadow(5) file instead; there must be a corresponding
       line in the /etc/shadow file, or else the user account is invalid.

       The encrypted password field may be empty, in which case no password is
       required to authenticate as the specified login name. However, some
       applications which read the /etc/passwd file may decide not to permit any
       access at all if the password field is blank.

       A password field which starts with an exclamation mark means that the password
       is locked. The remaining characters on the line represent the password field
       before the password was locked.
 Manual page passwd(5) line 1 (press h for help or q to quit)
```

Tilix: pccc@cis106: ~

1: pccc@cis106: ~

```
pccc@cis106:~$ cut -d":" -f1,6,7 --output-delimiter=">" /etc/passwd
root>/root>/bin/bash
daemon>/usr/sbin>/usr/sbin/nologin
bin>/bin>/usr/sbin/nologin
sys>/dev>/usr/sbin/nologin
sync>/bin>/bin/sync
games>/usr/games>/usr/sbin/nologin
man>/var/cache/man>/usr/sbin/nologin
lp>/var/spool/lpd>/usr/sbin/nologin
mail>/var/mail>/usr/sbin/nologin
news>/var/spool/news>/usr/sbin/nologin
uucp>/var/spool/uucp>/usr/sbin/nologin
proxy>/bin>/usr/sbin/nologin
www-data>/var/www>/usr/sbin/nologin
backup>/var/backups>/usr/sbin/nologin
list>/var/list>/usr/sbin/nologin
irc>/run/ircd>/usr/sbin/nologin
_apt>/nonexistent>/usr/sbin/nologin
nobody>/nonexistent>/usr/sbin/nologin
systemd-network>/>/usr/sbin/nologin
dhcpcd>/usr/lib/dhcpcd>/bin/false
tss>/var/lib/tpm>/bin/false
systemd-timesync>/>/usr/sbin/nologin
messagebus>/nonexistent>/usr/sbin/nologin
dnsmasq>/var/lib/misc>/usr/sbin/nologin
avahi>/run/avahi-daemon>/usr/sbin/nologin
speech-dispatcher>/run/speech-dispatcher>/bin/false
usbmux>/var/lib/usbmux>/usr/sbin/nologin
cups-pk-helper>/nonexistent>/usr/sbin/nologin
fwupd-refresh>/var/lib/fwupd>/usr/sbin/nologin
geoclue>/var/lib/geoclue>/usr/sbin/nologin
gnome-remote-desktop>/var/lib/gnome-remote-desktop>/usr/sbin/nologin
saned>/var/lib/saned>/usr/sbin/nologin
polkitd>/>/usr/sbin/nologin
rtkit>/proc>/usr/sbin/nologin
colord>/var/lib/colord>/usr/sbin/nologin
Debian-gdm>/var/lib/gdm3>/bin/false
```

2: pccc@cis106: ~

```
PASSWD(5)                File Formats and Configuration                PASSWD(5)

NAME
       passwd - the password file

DESCRIPTION
       /etc/passwd contains one line for each user account, with seven fields
       delimited by colons (":"). These fields are:

       •   login name

       •   optional encrypted password

       •   numerical user ID

       •   numerical group ID

       •   user name or comment field

       •   user home directory

       •   optional user command interpreter

       If the password field is a lower-case "x", then the encrypted password is
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       line in the /etc/shadow file, or else the user account is invalid.

       The encrypted password field may be empty, in which case no password is
       required to authenticate as the specified login name. However, some
       applications which read the /etc/passwd file may decide not to permit any
       access at all if the password field is blank.

       A password field which starts with an exclamation mark means that the password
       is locked. The remaining characters on the line represent the password field
       before the password was locked.
 Manual page passwd(5) line 1 (press h for help or q to quit)
```

Tilix: pccc@cis106: ~

1: pccc@cis106: ~

```
Lucky Charms
Maypo
Muesli Raisins, Dates, & Almonds
Muesli Raisins, Peaches, & Pecans
Mueslix Crispy Blend
Multi-Grain Cheerios
Nut&Honey Crunch
Nutri-Grain Almond-Raisin
Nutri-grain Wheat
Oatmeal Raisin Crisp
Post Nat. Raisin Bran
Product 19
Puffed Rice
Puffed Wheat
Quaker Oat Squares
Quaker Oatmeal
Raisin Bran
Raisin Nut Bran
Raisin Squares
Rice Chex
Rice Krispies
Shredded Wheat
Shredded Wheat 'n'Bran
Shredded Wheat spoon size
Smacks
Special K
Strawberry Fruit Wheats
Total Corn Flakes
Total Raisin Bran
Total Whole Grain
Triples
Trix
Wheat Chex
Wheaties
Wheaties Honey Gold
pccc@cis106:~$
```

```
pccc@cis106:~$ cut -d";" -f1,4 --output-delimiter=" calories =" ~/Documents/sample_files/Csv/cereal.csv
name calories =calories
String calories =Int
100% Bran calories =70
100% Natural Bran calories =120
All-Bran calories =70
All-Bran with Extra Fiber calories =50
Almond Delight calories =110
Apple Cinnamon Cheerios calories =110
Apple Jacks calories =110
Basic 4 calories =130
Bran Chex calories =90
Bran Flakes calories =90
Cap'n'Crunch calories =120
Cheerios calories =110
Cinnamon Toast Crunch calories =120
Clusters calories =110
Cocoa Puffs calories =110
Corn Chex calories =110
Corn Flakes calories =100
Corn Pops calories =110
Count Chocula calories =110
Cracklin' Oat Bran calories =110
Cream of Wheat (Quick) calories =100
Crispix calories =110
Crispy Wheat & Raisins calories =100
Double Chex calories =100
Froot Loops calories =110
Frosted Flakes calories =110
Frosted Mini-Wheats calories =100
Fruit & Fibre Dates, Walnuts, and Oats calories =120
Fruitful Bran calories =120
Fruity Pebbles calories =110
Golden Crisp calories =100
Golden Grahams calories =110
Grape Nuts Flakes calories =100
Grape-Nuts calories =110
```

Practice 3

```
Tilix: pccc@cis106: ~/Documents/sample_files/Csv
1: pccc@cis106: ~/Documents/sample_files/Csv
pccc@cis106:~/Documents/sample_files$ ls
Code  Config  Csv  Html  Json  Lst  Markdown  Rc  Toml  Txt  Xml  Yml
pccc@cis106:~/Documents/sample_files$ cd Csv
pccc@cis106:~/Documents/sample_files/Csv$ sort cars.csv
AMC Ambassador Brougham;13.0;8;360.0;175.0;3821.;11.0;73;US
AMC Ambassador DPL;15.0;8;390.0;190.0;3850.;8.5;70;US
AMC Ambassador SST;17.0;8;304.0;150.0;3672.;11.5;72;US
AMC Concord;19.4;6;232.0;90.00;3210.;17.2;78;US
AMC Concord;24.3;4;151.0;90.00;3003.;20.1;80;US
AMC Concord d/l;18.1;6;258.0;120.0;3410.;15.1;78;US
AMC Concord DL;23.0;4;151.0;0;3035.;20.5;82;US
AMC Concord DL 6;20.2;6;232.0;90.00;3265.;18.2;79;US
AMC Gremlin;18.0;6;232.0;100.0;2789.;15.0;73;US
AMC Gremlin;19.0;6;232.0;100.0;2634.;13.0;71;US
AMC Gremlin;20.0;6;232.0;100.0;2914.;16.0;75;US
AMC Gremlin;21.0;6;199.0;90.00;2648.;15.0;70;US
AMC Hornet;18.0;6;199.0;97.00;2774.;15.5;70;US
AMC Hornet;18.0;6;232.0;100.0;2945.;16.0;73;US
AMC Hornet;19.0;6;232.0;100.0;2901.;16.0;74;US
AMC Hornet;22.5;6;232.0;90.00;3085.;17.6;76;US
AMC Hornet Sportabout (sw);18.0;6;258.0;110.0;2962.;13.5;71;US
AMC Matador;14.0;8;304.0;150.0;3672.;11.5;73;US
AMC Matador;15.0;6;258.0;110.0;3730.;19.0;75;US
AMC Matador;15.5;8;304.0;120.0;3962.;13.9;76;US
AMC Matador;16.0;6;258.0;110.0;3632.;18.0;74;US
AMC Matador;18.0;6;232.0;100.0;3288.;15.5;71;US
AMC Matador (sw);14.0;8;304.0;150.0;4257.;15.5;74;US
AMC Matador (sw);15.0;8;304.0;150.0;3892.;12.5;72;US
AMC Pacer;19.0;6;232.0;90.00;3211.;17.0;75;US
AMC Pacer d/l;17.5;6;258.0;95.00;3193.;17.8;76;US
AMC Rebel SST;16.0;8;304.0;150.0;3433.;12.0;70;US
AMC Rebel SST (sw);0;8;360.0;175.0;3850.;11.0;70;US
AMC Spirit DL;27.4;4;121.0;80.00;2670.;15.0;79;US
Audi 100LS;20.0;4;114.0;91.00;2582.;14.0;73;Europe
Audi 100LS;23.0;4;115.0;95.00;2694.;15.0;75;Europe
Audi 100 LS;24.0;4;107.0;90.00;2430.;14.5;70;Europe
Audi 4000;34.3;4;97.00;78.00;2188.;15.8;80;Europe
```

```
Tilix: pccc@cis106: ~/Documents/sample_files/Csv
1: pccc@cis106: ~/Documents/sample_files/Csv
pccc@cis106:~/Documents/sample_files/Csv$ sort -o sorted_cars.csv cars.csv
pccc@cis106:~/Documents/sample_files/Csv$ ls
cars.csv  cereal.csv  contacts.csv  country.csv  fake_users.csv  sorted_cars.csv  users_and_ips.csv
pccc@cis106:~/Documents/sample_files/Csv$ wc -l sorted_cars.csv
408 sorted_cars.csv
pccc@cis106:~/Documents/sample_files/Csv$ wc -m sorted_cars.csv
22663 sorted_cars.csv
pccc@cis106:~/Documents/sample_files/Csv$
```

Question 2 Challenge

```
pccc@cis106:~$ wc -l access.log
100 access.log
pccc@cis106:~$ head -n 1 access.log
2025-03-18 17:35:41 Faye /dashboard SUCCESS
pccc@cis106:~$ tail -n 1 access.log
2025-08-23 18:26:18 Ein /settings FAIL
pccc@cis106:~$ head -n 15 access.log
2025-03-18 17:35:41 Faye /dashboard SUCCESS
2025-01-13 00:34:34 Shinji /settings DENIED
2025-07-22 14:49:03 Arisu /settings FAIL
2025-06-05 12:52:45 Asimov /upload FAIL
2025-02-05 14:06:38 Kei /settings FAIL
2025-04-08 10:34:22 Haruka /login FAIL
2025-01-15 13:58:04 Ein /settings SUCCESS
2025-10-15 11:11:48 Honami /logout SUCCESS
2025-09-16 15:15:09 Rize /upload SUCCESS
2025-12-16 15:05:25 Jet /settings SUCCESS
2025-01-19 09:28:34 Julia /dashboard DENIED
2025-06-10 13:21:22 Rei /404 DENIED
2025-12-03 22:06:04 Ein /logout FAIL
2025-08-06 05:20:05 Rei /reset DENIED
2025-04-08 13:10:13 Spike /reset SUCCESS
pccc@cis106:~$
```

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2025-03-18 17:35:41 Faye /dashboard SUCCESS
pccc@cis106:~$ tail -n 1 access.log
2025-08-23 18:26:18 Ein /settings FAIL
pccc@cis106:~$ head -n 15 access.log
2025-03-18 17:35:41 Faye /dashboard SUCCESS
2025-01-13 00:34:34 Shinji /settings DENIED
2025-07-22 14:49:03 Arisu /settings FAIL
2025-06-05 12:52:45 Asimov /upload FAIL
2025-02-05 14:06:38 Kei /settings FAIL
2025-04-08 10:34:22 Haruka /login FAIL
2025-01-15 13:58:04 Ein /settings SUCCESS
2025-10-15 11:11:48 Honami /logout SUCCESS
2025-09-16 15:15:09 Rize /upload SUCCESS
2025-12-16 15:05:25 Jet /settings SUCCESS
2025-01-19 09:28:34 Julia /dashboard DENIED
2025-06-10 13:21:22 Rei /404 DENIED
2025-12-03 22:06:04 Ein /logout FAIL
2025-08-06 05:20:05 Rei /reset DENIED
2025-04-08 13:10:13 Spike /reset SUCCESS
pccc@cis106:~$ tail -n 15 access.log
2025-03-28 18:27:45 Shinji /download DENIED
2025-03-06 15:10:50 Spike /logout DENIED
2025-07-21 19:52:51 Annie /download FAIL
2025-09-18 03:30:50 Spike /404 DENIED
2025-10-15 23:44:37 Asimov /download DENIED
2025-04-18 15:03:19 Ein /logout SUCCESS
2025-06-19 08:12:33 Arisu /download SUCCESS
2025-03-11 18:09:29 Haruka /upload FAIL
2025-09-23 10:44:48 Seras /upload FAIL
2025-11-23 08:59:52 Spike /download FAIL
2025-01-18 12:59:35 Rize /reset FAIL
2025-10-03 13:40:08 Spike /logout DENIED
2025-11-27 06:29:49 Eren /download DENIED
2025-05-21 19:29:06 Edward /admin SUCCESS
2025-08-23 18:26:18 Ein /settings FAIL
pccc@cis106:~$
```

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2025-05-21 19:29:06 Edward /admin SUCCESS
2025-08-23 18:26:18 Ein /settings FAIL
pccc@cis106:~$ cut -d " " -f3 access.log
Faye
Shinji
Arisu
Asimov
Kei
Haruka
Ein
Honami
Rize
Jet
Julia
Rei
Ein
Rei
Spike
Shinji
Ken
Ken
Edward
Ein
Rize
Rize
Seras
Seras
Spike
Julia
Eren
Annie
Asimov
Asimov
Edward
Julia
Touka
Edward
```

```
pccc@cis106:~$ sort -k3 access.log
2025-07-19 08:41:39 Annie /dashboard SUCCESS
2025-07-21 19:52:51 Annie /download FAIL
2025-11-05 11:39:10 Annie /login DENIED
2025-02-12 08:10:27 Arisu /404 DENIED
2025-06-26 17:46:28 Arisu /dashboard FAIL
2025-06-27 22:09:43 Arisu /dashboard FAIL
2025-06-19 08:12:33 Arisu /download SUCCESS
2025-07-22 14:49:03 Arisu /settings FAIL
2025-10-15 23:44:37 Asimov /download DENIED
2025-07-11 14:06:14 Asimov /login DENIED
2025-07-27 09:40:49 Asimov /login SUCCESS
2025-06-05 12:52:45 Asimov /upload FAIL
2025-02-07 01:10:58 Asimov /upload SUCCESS
2025-07-07 07:12:56 Edward /404 DENIED
2025-08-18 08:51:04 Edward /404 FAIL
2025-05-21 19:29:06 Edward /admin SUCCESS
2025-08-19 18:56:59 Edward /dashboard DENIED
2025-01-21 15:55:38 Edward /download DENIED
2025-12-07 06:59:34 Edward /reset SUCCESS
2025-10-20 00:04:20 Edward /settings SUCCESS
2025-11-28 07:02:48 Edward /settings SUCCESS
2025-01-05 00:08:08 Ein /404 DENIED
2025-12-03 22:06:04 Ein /logout FAIL
2025-04-18 15:03:19 Ein /logout SUCCESS
2025-05-08 13:40:17 Ein /reset DENIED
2025-08-17 21:38:54 Ein /reset FAIL
2025-11-11 14:40:13 Ein /settings DENIED
2025-08-23 18:26:18 Ein /settings FAIL
2025-01-15 13:58:04 Ein /settings SUCCESS
2025-08-23 16:24:27 Ein /upload DENIED
2025-01-14 09:32:47 Eren /admin DENIED
2025-07-14 07:00:47 Eren /admin FAIL
2025-11-27 06:29:49 Eren /download DENIED
2025-05-08 22:35:14 Eren /download FAIL
2025-04-24 14:12:29 Eren /logout SUCCESS
2025-05-23 09:14:07 Faye /admin SUCCESS
```

```
pccc@cis106:~$ sort -k4 access.log
2025-01-05 00:08:08 Ein /404 DENIED
2025-02-12 08:10:27 Arisu /404 DENIED
2025-06-10 13:21:22 Rei /404 DENIED
2025-07-07 07:12:56 Edward /404 DENIED
2025-09-18 03:30:50 Spike /404 DENIED
2025-01-15 09:10:53 Seras /404 FAIL
2025-08-18 08:51:04 Edward /404 FAIL
2025-11-17 18:39:23 Spike /404 FAIL
2025-11-26 01:45:53 Julia /404 FAIL
2025-04-01 21:59:06 Julia /404 SUCCESS
2025-04-24 19:54:08 Rei /404 SUCCESS
2025-11-21 16:17:28 Seras /404 SUCCESS
2025-01-14 09:32:47 Eren /admin DENIED
2025-02-08 12:02:42 Rize /admin DENIED
2025-06-02 11:48:34 Honami /admin DENIED
2025-02-14 02:52:06 Julia /admin FAIL
2025-07-14 07:00:47 Eren /admin FAIL
2025-07-17 09:11:26 Kei /admin FAIL
2025-12-26 14:05:42 Spike /admin FAIL
2025-04-12 08:37:29 Seras /admin SUCCESS
2025-05-21 19:29:06 Edward /admin SUCCESS
2025-05-23 09:14:07 Faye /admin SUCCESS
2025-01-19 09:28:34 Julia /dashboard DENIED
2025-08-19 18:56:59 Edward /dashboard DENIED
2025-11-14 07:12:03 Faye /dashboard DENIED
2025-12-26 18:36:58 Shinji /dashboard DENIED
2025-06-26 17:46:28 Arisu /dashboard FAIL
2025-06-27 22:09:43 Arisu /dashboard FAIL
2025-08-03 02:12:55 Ken /dashboard FAIL
2025-03-18 17:35:41 Faye /dashboard SUCCESS
2025-03-20 09:16:38 Ken /dashboard SUCCESS
2025-03-28 06:51:19 Spike /dashboard SUCCESS
2025-07-19 08:41:39 Annie /dashboard SUCCESS
2025-08-11 18:51:51 Honami /dashboard SUCCESS
2025-11-19 18:39:41 Honami /dashboard SUCCESS
2025-01-04 09:38:27 Honami /download DENIED
```

```
Edward /admin
Ein /settings
pccc@cis106:~$ cut -d ' ' -f3,4 --output-delimiter=' access: ' access.log
Faye access: /dashboard
Shinji access: /settings
Arisu access: /settings
Asimov access: /upload
Kei access: /settings
Haruka access: /login
Ein access: /settings
Honami access: /logout
Rize access: /upload
Jet access: /settings
Julia access: /dashboard
Rei access: /404
Ein access: /logout
Rei access: /reset
Spike access: /reset
Shinji access: /dashboard
Ken access: /dashboard
Ken access: /dashboard
Edward access: /404
Ein access: /upload
Rize access: /logout
Rize access: /upload
Seras access: /login
Seras access: /404
Spike access: /admin
Julia access: /upload
Eren access: /admin
Annie access: /dashboard
Asimov access: /upload
Asimov access: /login
Edward access: /settings
Julia access: /404
Touka access: /settings
Edward access: /download
```